Unsealed roads: a fine sediment source for the Reef

The source

Erosion of unsealed roads in Queensland sends excess fine sediment downstream to the Great Barrier Reef (GBR) lagoon.

The harm

Sediment cuts light, buries or smothers corals and seagrasses, increases disease and alters microbial communities in a World Heritage-listed ecosystem.

The evidence gap

No known published studies cover the costs or effectiveness of road management for controlling fine sediment loss in GBR catchments.

A · July 2021
before works



B · Nov 2021
after DRFA reconstruction



C · Dec 2021
during a minor storm



D · June 2023
after an average wet season



E · Dec 2023
after Cyclone Jasper



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The task

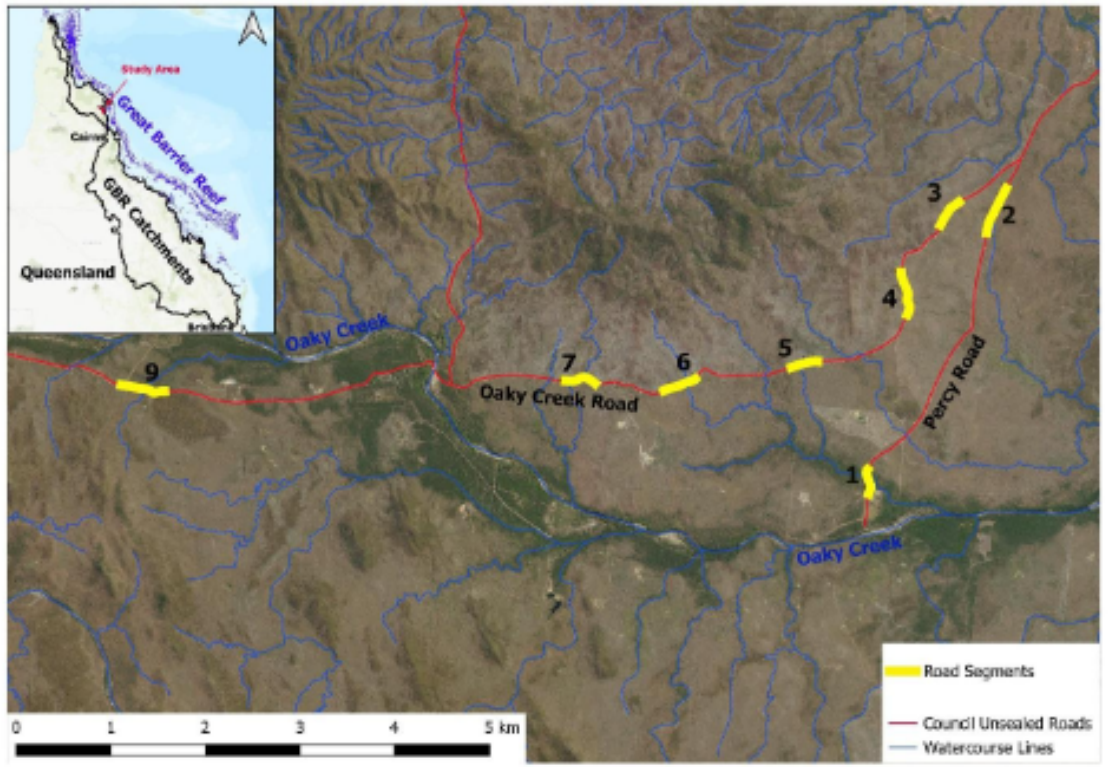
Background
South Cape York Catchments (SCYC) modelled erosion on Percy and Oaky Creek Road segments.

Scenarios
Four maintenance options, from the status quo to a full bitumen seal.

Objective
Identify the most effective and feasible scenario from these case studies.

Why it matters
Evidence for local, State and Federal governments to invest in road betterments.

This project
An economic analysis comparing the scenarios on environmental, economic and social outcomes.



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Road management scenarios

SCENARIO 1 Business-as-usual · *base case*

Routine council practice: grading to reshape the road crown, gravel re-sheeting, cutting and re-cutting table drains and cut-off v-drains, and full vegetation removal along batters and verges. Disaster damage repaired like-for-like.

SCENARIO 2 Alternative maintenance · *no capital works*

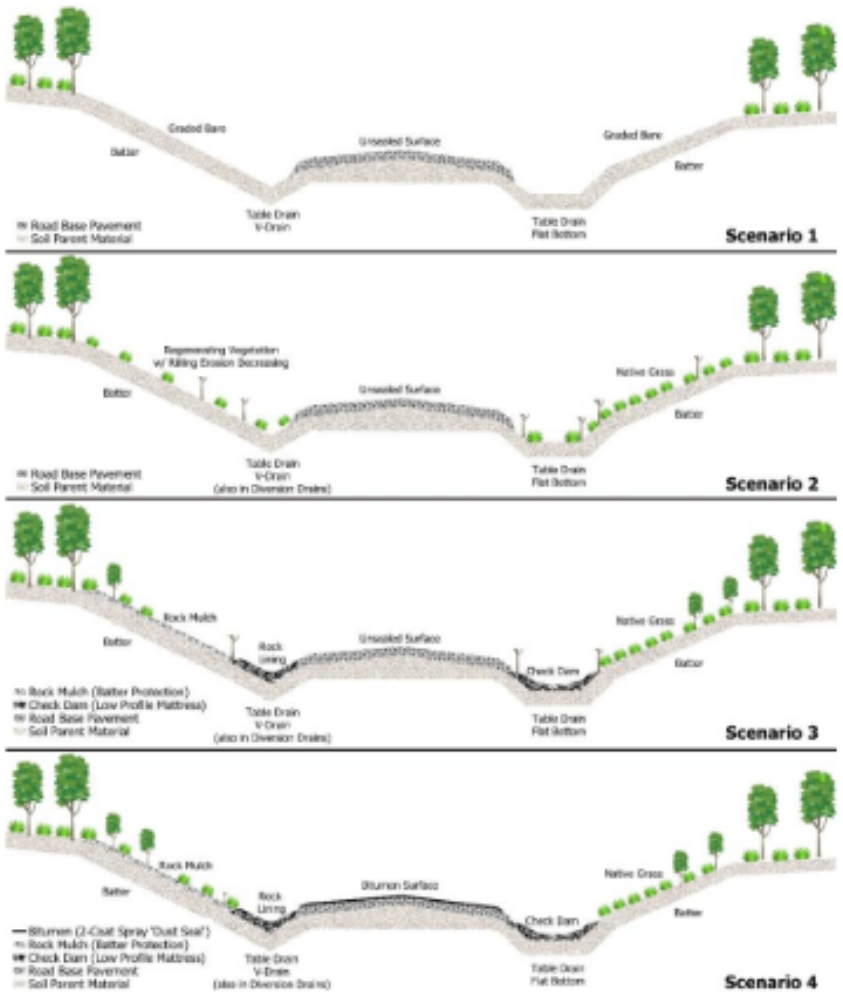
Stop repeated grading of table drains, cut-off drains, verges and batters; use selective herbicide to prevent tree regrowth; spot machinery maintenance only on drains with erosion problems.

SCENARIO 3 Major erosion control · *builds on Scenario 2*

Scenario 2 plus capital works: soil stabilisation on eroding batters and drains, rock mulch and gully rock chutes, and rock or concrete bed-level floodways at stream crossings.

SCENARIO 4 Full betterment · *builds on Scenario 3*

Scenario 3 plus a bitumen dust seal over the full 21.5 km – a dust seal of the existing alignment, not an upgrade to a sealed road. Reseal required after 20 years.



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Four road management scenarios delivered different erosion outcomes

1

Business-as-usual

Graders reshape the road crown, add gravel material, install and re-cut table drains and cut-off v-drains, and remove all vegetation along road batters and verges.

55.3 t/km/yr

2

Alternative maintenance

Cease repeated grading of table drains, cut-off drains, verges and batters. Selective herbicide vegetation management prevents tree regrowth, with spot machinery maintenance of drains with erosion problems.

46.5 t/km/yr

3

Major erosion control

Everything in Scenario 2, plus capital improvements to stabilise soil on eroding batters and drains, and rock or concrete bed-level floodways at stream crossings.

31.7 t/km/yr

4

Full betterment

Everything in Scenario 3, plus a bitumen dust seal of the road surface. This is a dust seal of the existing alignment, not a full upgrade to a sealed road.

17.6 t/km/yr

Where the benefits come from

Every option is compared on cost. So a "benefit" here means money saved and impact avoided compared with carrying on as usual (Scenario 1).

Less soil washing into the Reef

Grading, and drains that wash out, send fine soil into creeks that flow to the Great Barrier Reef. Cleaning that up elsewhere costs money, so soil we don't lose is money saved. We measured the erosion with 3D laser scans and priced it using what real Reef clean-up projects cost.

A smaller yearly repair bill

Grading the same drains and verges repeatedly is expensive. If we investment in maintenance that avoids having to do so, we save money for ongoing maintenance.

Other benefits

Weeds and biosecurity · Biodiversity · Wildlife · Road safety · Road closures · Wear and tear on vehicles

These benefits are hard to value without additional data but could be critical. How would these possibly impact the results?

Information provided

- Cook Shire Council have provided information on the costs of different road maintenance scenarios, SCYC have provided the data on modelled erosion rates.
- **Research is required to quantify value of avoided cost from fine sediment transport to the GBR.**

THE PROBLEM

Sediment has no market price

Nobody buys or sells a tonne of fine sediment in the Reef lagoon, so there is no price to look up.

THE PROXY

Borrow a price from a real program

Governments already pay to keep sediment out of the Reef. What they pay per tonne stands in for the price.

THE PROGRAM USED

Reef Trust gully control

Reef Trust spending shows what it costs to stop a tonne of sediment somewhere else.

Your task

To prepare a report on the different road maintenance scenarios for SCYC.

Introduction

- Overview of issues with existing road maintenance
- What are the implications of the current approaches?
- What is the purpose of this project?

Cost-effectiveness analysis

- Compare the scenarios by lifecycle cost per tonne sediment abated

Cost-benefit analysis

- Identify and quantify the key benefits of different scenarios against the base case
- Calculate the NPV

Conclusion

- Summarise findings and provide a recommendation based on CEA and CBA results
- Highlight any tradeoffs, sources of uncertainties and implications for long-term success



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